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The application of genetic algorithm and fuzzy evaluation method in Teachers' performance appraisal

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Abstract: ***Research on the application of genetic algorithm and fuzzy evaluation method in Teachers' performance appraisal? Genetic algorithm (GA) is a search technique that uses heuristic rules to solve problems. It has been widely used in artificial intelligence, engineering, economics, finance and medicine. GA also evaluates teacher performance evaluation by using fuzzy logic system to evaluate teacher performance. In this research work, we developed a GA to find the best teacher from all available candidates according to the test results and evaluate them using a fuzzy logic system. This is based on the concept that teachers evaluate students' academic performance. We propose an automatic system that uses fuzzy logic as a tool to evaluate teachers at different stages of the educational process through multiple measurement methods such as scores, examination scores, and final examination scores.***

Keyword: Genetic algorithm; Fuzzy evaluation; teacher; performance appraisal

I. INTRODUCTION

Teacher performance assessment is the central link of human resource management in schools, which provides the basis and basis for teachers' salary, promotion and professional title evaluation. High quality teachers are the direct driving force for the development of colleges and universities, and a perfect and scientific performance evaluation system for college teachers is an important guarantee to promote the development of colleges and universities. The evaluation of teachers' teaching quality is an important part of the content of teachers' performance evaluation in Colleges and universities. It plays an active role in guiding teachers to improve teaching level and teaching methods.

How to build a teaching quality index system is an important guide for the quality of education and teaching. It is the key to the success of teaching quality evaluation. The index system not only represents the specific assessment scope proposed by the school for the purpose of assessment, but also points out the requirements of the school on the teaching ability of teachers[1]. The index weight specifies the degree of attention in a certain aspect.

Colleges and universities are the cradle of elite talents, the birthplace of advanced knowledge, and also occupy the highland of human resources. The level of teaching quality is directly related to the quality of higher education. Teachers are the direct driver of improving teaching quality. High-level teachers not only represent the level of the school, but also the guide of talents. An excellent teacher team depends on the importance of human resources management by the majority of colleges and universities. The cultivation of innovative talents is the primary task of the construction of colleges and universities. Therefore, colleges and universities must explore human resources management strategies that have guiding significance for their own human resources development and management, and ultimately achieve the improvement of organizational performance[2]. Therefore, it is of great significance to do a good job in the performance management of college teachers.

The performance management of teachers is the focus of human resources management. The so-called teacher performance refers to the performance and effectiveness of teachers. Generally speaking, it includes two aspects: first, the quality and behavior of teachers consistent with the requirements of schools; The second is the result of teachers' work, which is reflected in the quantity, quality and impact of the work completed. Different from other work, the content of teachers' performance assessment is also very broad. The assessment of teachers' performance requires comprehensive consideration of the school's own teaching development objectives, the jurisdiction and scope of authority of each department, the characteristics of teachers and students in the school, the indicator assessment system, the assessment standards and assessment methods[3]. The purpose of the assessment is to improve the teachers' teaching satisfaction, make them "do their best" and "make the best use of their talents", make the teachers' performance management closely cooperate with the school's strategic objectives, so as to facilitate the sustainable development of the teachers' team in the fierce competitive environment and enhance the comprehensive teaching strength of the school. So we propose the overall frame in figure 1.

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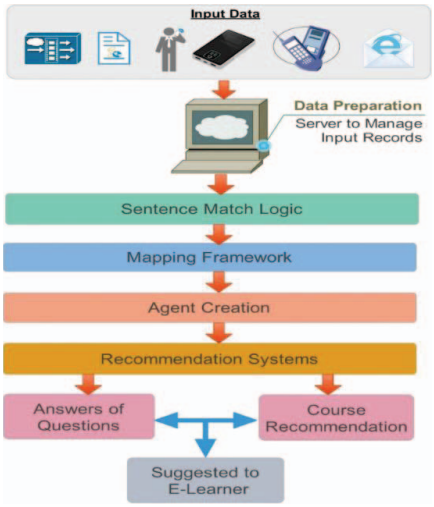


Figure 1. The overall frame chart

II. RELATED WORK

A.fuzzy evaluation

Apply the related theory of membership degree to transform the subjective evaluation into quantitative evaluation, and make an overall evaluation of the things that contain multiple attributes or are affected by multiple factors. In the application of this method in the performance appraisal, we should first get the more subjective data through the evaluation, then use the fuzzy theory to transform the obtained data, eliminate the error caused by subjective judgment, and finally get the comprehensive evaluation result.

The fuzzy evaluation method can be divided into the following steps:

- Establish factor set and comment set: factor set refers to the evaluation items that need to be established for the evaluation object. The judges give an evaluation grade on these evaluations according to the performance of the evaluation object in the actual work, such as excellent, good, allowable, poor, etc. the set of these evaluation grades is called comment set.
- Single factor evaluation: combined with the analysis of analytic hierarchy process, single factor evaluation usually starts from the object level of the evaluation set, evaluates a factor and determines the evaluation level. The fuzzy evaluation matrix can be obtained by combining the evaluation results of all the auxiliary factors under a factor.
- Comprehensive evaluation: the final evaluation result is obtained by comprehensively considering all factors of the evaluation object, and the evaluation matrix obtained by single factor evaluation is processed with fuzzy comprehensive evaluation operator.

In addition, the fuzzy comprehensive evaluation method is often used in combination with the analytic hierarchy process. The steps of the algorithm are to obtain the weight vector of the index system through the analytic hierarchy process, and then get the final evaluation result through the evaluation by the judges[4].

Generally, analytic hierarchy process (AHP) and fuzzy comprehensive evaluation method need to be mixed. The steps of the algorithm are generally to obtain the weight vector of the index system through AHP, then score by the judges, and then obtain the final evaluation result through the fuzzy comprehensive evaluation algorithm.

B.principles of teacher performance management

The optimization design principle of the performance appraisal management system is mainly based on the school development strategy and guided by the improvement of teachers' performance. The specific principles are as follows:

- Principle of combining qualitative evaluation with quantitative evaluation

Qualitative evaluation mainly refers to the subjective evaluation made by the evaluator according to the formulated evaluation content and his own observation on the evaluated. Sometimes, the evaluator uses excellent, good, qualified and unqualified to distinguish several grades. Sometimes, the evaluator writes comments directly. The qualitative evaluation pays more attention to the development of "quality". Quantitative evaluation mainly uses the method of scoring for accurate evaluation, which is relatively objective[5]. However, some qualities and behaviors without measurability cannot be evaluated. Therefore, teachers' performance evaluation needs to combine qualitative evaluation and quantitative evaluation, both of which are indispensable.

- Principle of difference

On the one hand, the principle of diversity means that the assessment level of teachers with different professional bases is different, and different assessment methods should be adopted for teachers with different professional bases to meet the development needs of teachers at various stages; On the other hand, the evaluation of teachers can't be all regular, and mandatory classification can be carried out when necessary. The different evaluation results can provide reference for teachers' promotion or personnel deci-

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sion-making, and will also intensify teachers' work enthusiasm and work potential.

When formulating the assessment contents, the school's strategic development objectives shall be taken as the guidance to ensure that the assessment indicators are scientific, avoid human subjectivity, and the assessment standards are correct to the matter and not to the person. In the assessment process, give play to the mass supervision function, make the operation process public, and the assessor shall explain the part with doubts to ensure the fairness of the assessment results.

III. RESEARCH ON THE APPLICATION OF GENETIC ALGORITHM AND FUZZY EVALUATION METHOD IN TEACHERS' PERFORMANCE APPRAISAL

The evaluation index set is a collection of evaluation influence factors obtained from different perspectives of the examinees. Generally, teachers and professionals in relevant universities participate in discussion and investigation according to their own characteristics in the teaching process and finally establish it[6]. The indicator set is established according to the typical three-layer structure (target layer, criterion layer and object layer) of the hierarchy, for example:

$$u_{k+1}(t) = u_k(t) + \Gamma_{l1}\dot{e}_k(t) + \Gamma_{l2}\dot{e}_{k+1}(t) + \Gamma_{p1}\Delta\dot{e}_k(t) + \Gamma_{p2}\Delta\dot{e}_{k+1}(t) \quad (1)$$

According to the established index system, the analytic hierarchy process is applied to solve the weight value of each index, as follows:

Construct judgment matrix of index system

Taking the judgment matrix B of the criterion layer as an example, the value of each element in the matrix is the ratio of the criterion layer index represented by the label of the row where the element is located to the importance of the criterion layer index represented by the label of the column where the element is located with respect to the previous layer, that is, the target layer. Psychology shows that most people's ability to distinguish things is between 5 and 9 levels. Therefore, the 1-9 scale method can be widely used to distinguish the degree of personal judgment on things. Because this method is simple and convenient, it can be used by professional and non professional personnel.

Assuming that the number of first-class indicators in the teaching quality indicator system is n, a judgment matrix between the criterion level and the target level is established. See formula 2:

$$Q = (E(t) - M_2C), \left(\Gamma_{p1}C + M_1C + M_2C\right)\dot{x}_d(t) = Z, \Gamma_{p1}C = F \quad (2)$$

The value of the main diagonal of the judgment matrix is the ratio of each index to itself, i.e. 1. The symmetrical elements of the upper triangle and the lower triangle with respect to the main diagonal are reciprocal to each other. Therefore, it is only necessary to calculate the value of the upper triangle element, and the total number of comparisons between two is n (n-1) / 2[7].

In the process of applying the fuzzy comprehensive evaluation method, the evaluator only needs to make a qualitative conclusion on the level attribution of the relevant indicators of the evaluator to obtain a more accurate evaluation result, which can greatly improve the work efficiency and provide an effective guarantee for the stability of the evaluation process and the fairness of the evaluation results.

IV. SIMULATION ANALYSIS

In order to get the weight value of each index, the judgment matrix in analytic hierarchy process must be solved. The judgment matrix is an important operation object of the whole algorithm. The quality of the judgment matrix directly affects the objectivity and rationality of the index weight. Therefore, after the eigenvector is obtained through the judgment matrix, the consistency test is also carried out[8].

In the practical application of AHP, since the method of constructing the judgment matrix is to give each element value in a subjective way, it is difficult to achieve complete consistency of the judgment matrix. When the consistency of a judgment matrix of a certain layer is poor, it is necessary to reset all the element values of the matrix until the consistency of the judgment matrix meets the conditions. This repeated reset process makes the application of analytic hierarchy process very cumbersome and its practicability greatly reduced[9]. Therefore, it is very necessary and key to improve the analytic hierarchy process (AHP) so that the system can efficiently and quickly find a more reasonable judgment matrix. The following figure 2 shows the weight value evaluation process of each indicator.

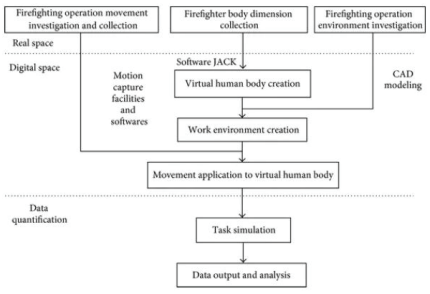


Fig. 2. Weight value evaluation process

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For different evaluated objects or different types of judges, the evaluation membership function matrix will be different. In order to verify the feasibility of the above algorithm, this section only lists and analyzes the process of the evaluation of 8 teachers in a middle school in the actual project. Determine the evaluation factor set and indicator weight. According to the analysis results of the indicator system of performance appraisal using AHP in Chapter 3, it can be known that the five indicators of the criteria layer can form the evaluation factor set of the criteria layer, and the object layer can form the evaluation factor set of the five object layers according to the five indicators of the criteria layer[10].

According to the above, the normalized weight vectors of each evaluation factor set can be obtained. The comparison of the weight vector curves of the evaluation factor sets of the criteria layer is shown in Figure 3 and Figure 4:

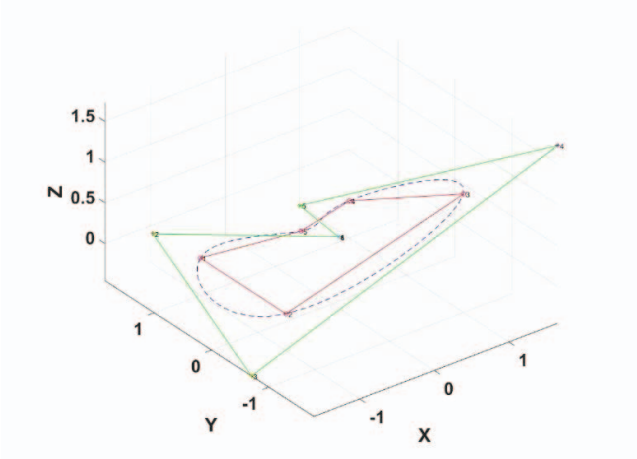


Fig. 3. Comparison of weight vector curves of evaluation factor sets

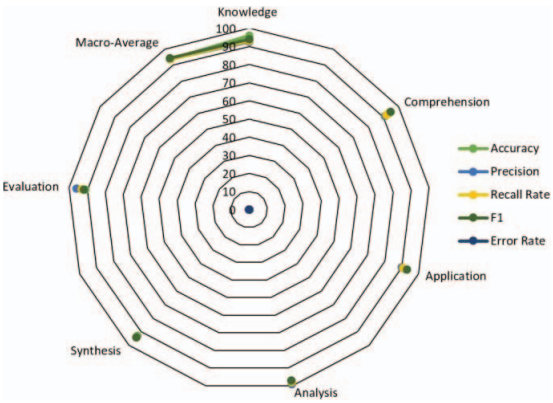


Fig. 4. Evaluation effect

V. CONCLUSION

The main purpose of this study is to find out whether the use of genetic algorithms and fuzzy evaluation methods can improve the performance evaluation process of teachers, and give more reliable results than those obtained by traditional methods such as descriptive statistics or visual inspection. In this study, we use fuzzy evaluation method to evaluate the performance of teachers in different levels of educational institutions. The results show that the fuzzy evaluation method can give more accurate results than other methods such as genetic algorithm and neural network.

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